

1:-

V4P: HBM BW = $1.2e12$

- FLOPs/s hf16 = $2.75e14$
 - HBM capacity = 32 GB
 - 200 Billion = $200 \cdot 10^9$ (hf16)
 - Split across 32 TPUs V4P.
 - Bytes per TPU = $\frac{200 \cdot 10^9 \cdot 2}{32}$
 $= \frac{400 \cdot 10^9}{32} = 12.5 \cdot 10^9 \text{ Bytes} = 12.5 \text{ GB}$
- $12.5 \text{ GB} \leq 32 \text{ GB}$. So it does fit.

- Each TPU HBM has 12.5 GB of weights to send to the Tensor Cores:

$$\Rightarrow T_{\text{comm}} = \frac{12.5 \cdot 10^4 \text{ Bytes}}{1.2 \cdot 10^{12} \text{ Bytes/s}}$$

$$\approx 10 \cdot 10^{-3} \text{ s} = 10 \text{ ms.}$$

• Question 2:

- TPU vSe pod :

• Pod size : 10×16

• Host size : 4×2

• HBM per chip : 16 GB

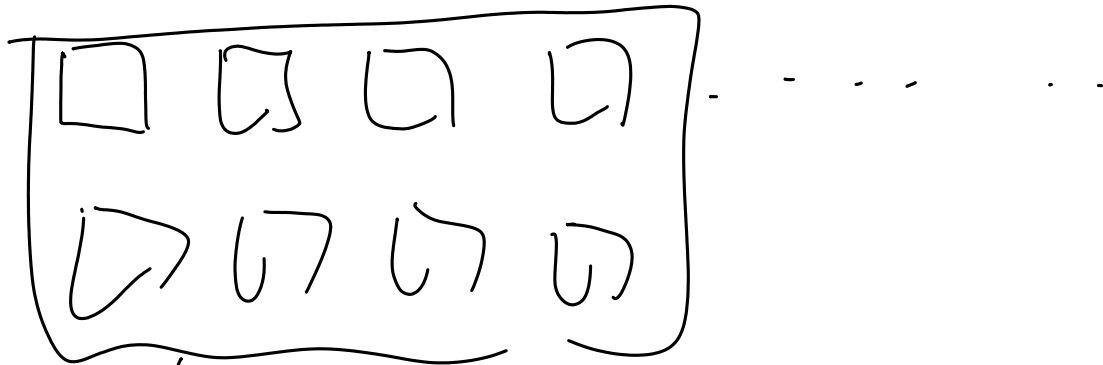
• FLOPs $_{\text{Hf16}}$

per chip =

$$1.97 \cdot 10^{14}$$

- Host size = $2 \times 4 = 8$

- 16×16 how many blocks of 2×4 fit:



One Host

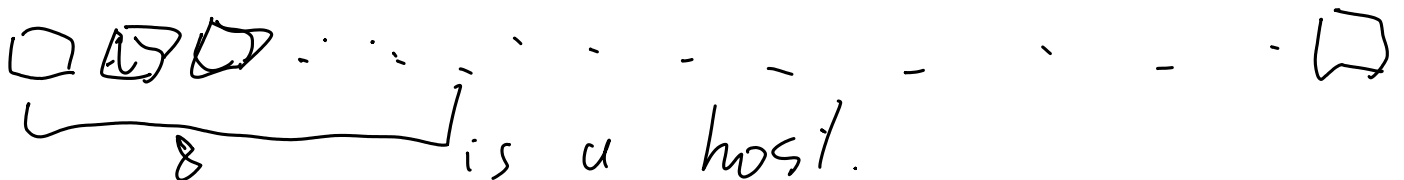
- 4 hosts to cover one "row"

- 8 hosts to cover one "column"

$\Rightarrow 32$ hosts.

- I could also just flatten:

$$16 \times 16 = 256$$



- TPU v2e has only one Tensor Core per chip.

~~256~~ 256 Tensor Core.

- Also you can derive by cores per host and from the book you see using previous result:

$$32 \times 8 = 256 \text{ Tensor Core}$$

Hosts ✓
in the pod. ↓
cores per host.

- Total FLOP/s = $256 \times 1.97 \times 10^{14}$
 $\approx 500 \times 10^{14}$.

- Total HBM = $256 \times 16 \cdot 10^9 = 4096 \times 10^9$
 $= 4096 \text{ GB}$.

• TPU v5P pod:

• $16 \times 20 \times 28$ Pod size

• $2 \times 2 \times 1$ Host size

• Flatten $16 \times 20 \times 28 = 8960$ chips.

• Total Hosts = $\frac{8960}{4} = 2240$

• Tensor Cores = $8960 \times 2 = 17920$

• Total HBM = $8960 \times 96 \cdot 10^9$

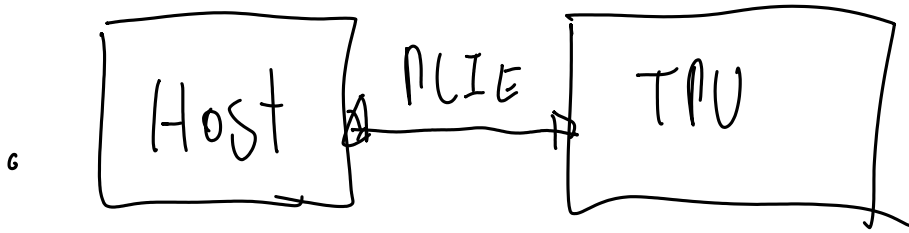
= $860160 \cdot 10^9 \approx 860 \cdot 10^3 \cdot 10^9$

= 860 TB.

• Total FLOPS/s = $8960 \times 4.54 \cdot 10^{14}$

$\approx 41000 \times 10^{14} \approx 4.1 \text{ Exa FLOPS/s}$

• Question 3:



• TPU v6e

• $bf16 [B, D] \rightarrow bf16 [D, F] \rightarrow bf16 [B, F]$

• $B = 40, F = 40$

• PCIe BW = 1.6×10 bytes/s

• FLOPS/s = 9.20×14

• Assuming I can overlap communication with compute:

$$A_i = \frac{2 \cdot B \cdot D \cdot F}{2BD + 2 \cdot DF + 2 \cdot BF} > \frac{9.20 \cdot 10^{14}}{1.6 \cdot 10^{10}}$$

$$A_i \approx \frac{\cancel{2} \cdot B \cdot D \cdot \cancel{4} D}{2 \cdot D \cdot \cancel{4} \cdot D} = \frac{B \cdot \cancel{D^2}}{\cancel{D^2}}$$

$$\Rightarrow B > 5.75 \cdot 10^4 = 57.5 \cdot 10^3$$

like 57 sequences of 1024 seq/len:

• Question 4:

• VSe • HMM BW = $8.2e11$

• FLOPs/s int8 = $3.94e14$

- $VMEM\ BW \approx HAM\ BW \times 22$

$$\approx 8.2 \cdot 10^{11} \times 22 = 180.4 \times 10^{11}$$

Bytes / s.

- $int8[0, 4096] @ int8[4096, 16384]$
 $\rightarrow int8[0, 16384]$

- $T = \max(T_{comms}, T_{math})$

- Note : $16384 = 4 \cdot 4096$

- So let $D = 4096$ and $F = 4096 \cdot 4$

$\Rightarrow F = 4 \cdot D$ as before.

$$[B, D] \cdot [D, 4 \cdot D] \rightarrow [B, 4D]$$

$$\Rightarrow T\text{-comm} = \frac{B \cdot D + 4 \cdot D^2 + B \cdot 4 \cdot D}{8.2 \text{ e } 11}$$

$$T\text{-comm} = \frac{B \cdot (D + 4 \cdot D) + 4 \cdot D^2}{8.2 \text{ e } 11}$$

$$= \frac{B \cdot (5D) + 4 \cdot D^2}{8.2 \text{ e } 11}$$

$$\bullet T_{\text{math}} = \frac{2 \cdot B \cdot D \cdot F}{3.94 \cdot 10^{14}} = \frac{8 \cdot B \cdot D^2}{3.94 \cdot 10^{14}}$$

$$\bullet T_{\text{operation}} = \max \left(T_{\text{comms}}(B), T_{\text{math}}(B) \right)$$

$$= \max \left(\frac{B \cdot 5 \cdot D + 4 \cdot D^2}{8.2 \cdot 10^{11}}, \frac{8 \cdot B \cdot D^2}{3.94 \cdot 10^{14}} \right)$$

$$\Rightarrow T_{\text{operation}}(B, BW)$$

$$= \max \left(\frac{B \cdot 5 \cdot D + 4 \cdot D^2}{BW}, \frac{8 \cdot B \cdot D^2}{3.94 \cdot 10^{14}} \right)$$

Since:

$$\text{VMEM} : 128 \text{ MiB} = 128 \cdot 10^6$$

$$4096 \times (4 \cdot 4096) \approx 68 \cdot 10^6$$

at least the weights do fit
in VMEM.

• Bottleneck: (compute bound when?)

$$\bullet T_{\text{comms}}(B, BW) \leq T_{\text{math}}(B)$$

$$\Rightarrow \frac{B \cdot 5 \cdot D + 4 \cdot D^2}{BW} \leq \frac{8 \cdot B \cdot D^2}{3.94 \cdot 10^{14} \text{ FLOPS/s}}$$

$$\Rightarrow \frac{B \cdot 5 \cdot D + 4 \cdot D^2}{8 \cdot B \cdot D^2} \leq \frac{BW}{3.94 \cdot 10^{14}} = (AI)^{-1}$$

$$\Rightarrow \frac{B \cdot 5 \cdot D}{8 \cdot B \cdot D^2} + \frac{4 \cdot D^2}{8 \cdot B \cdot D^2} \leq (AI)^{-1}$$

$$\Rightarrow 0 \leq \frac{5}{8 \cdot 10} + \frac{4}{8 \cdot 10} \leq (AI)^{-1}$$

$$\Rightarrow \frac{4}{8 \cdot 10} \leq (AI)^{-1} - \frac{5}{8 \cdot 10}$$

$$\Rightarrow \frac{1}{2 \cdot B} \leq (AI)^{-1} - \frac{5}{8 \cdot 10}$$

$$\Rightarrow \frac{1}{B} \leq 2 \cdot (AI)^{-1} - \frac{10}{8 \cdot 10}$$

$$\Rightarrow B \geq \frac{1}{2 \cdot \left(\frac{1}{AI} \right) - \frac{10}{8 \cdot 10}}$$

$$\Rightarrow B \geq \frac{1}{2 \cdot \left(\frac{BW}{3.94 \cdot 10^4} \right) - 3 \cdot 10^{-4}}$$

$$\cdot BW_{HBM} = 8.2 \text{ e } 11$$

$$\cdot BW_{VME} = 180.4 \text{ e } 11$$

$$\Rightarrow B \geq \frac{1}{2 \cdot \frac{8.2 \text{ e } 11}{3.94 \text{ e } 14} - 3 \cdot 10^{-4}}$$

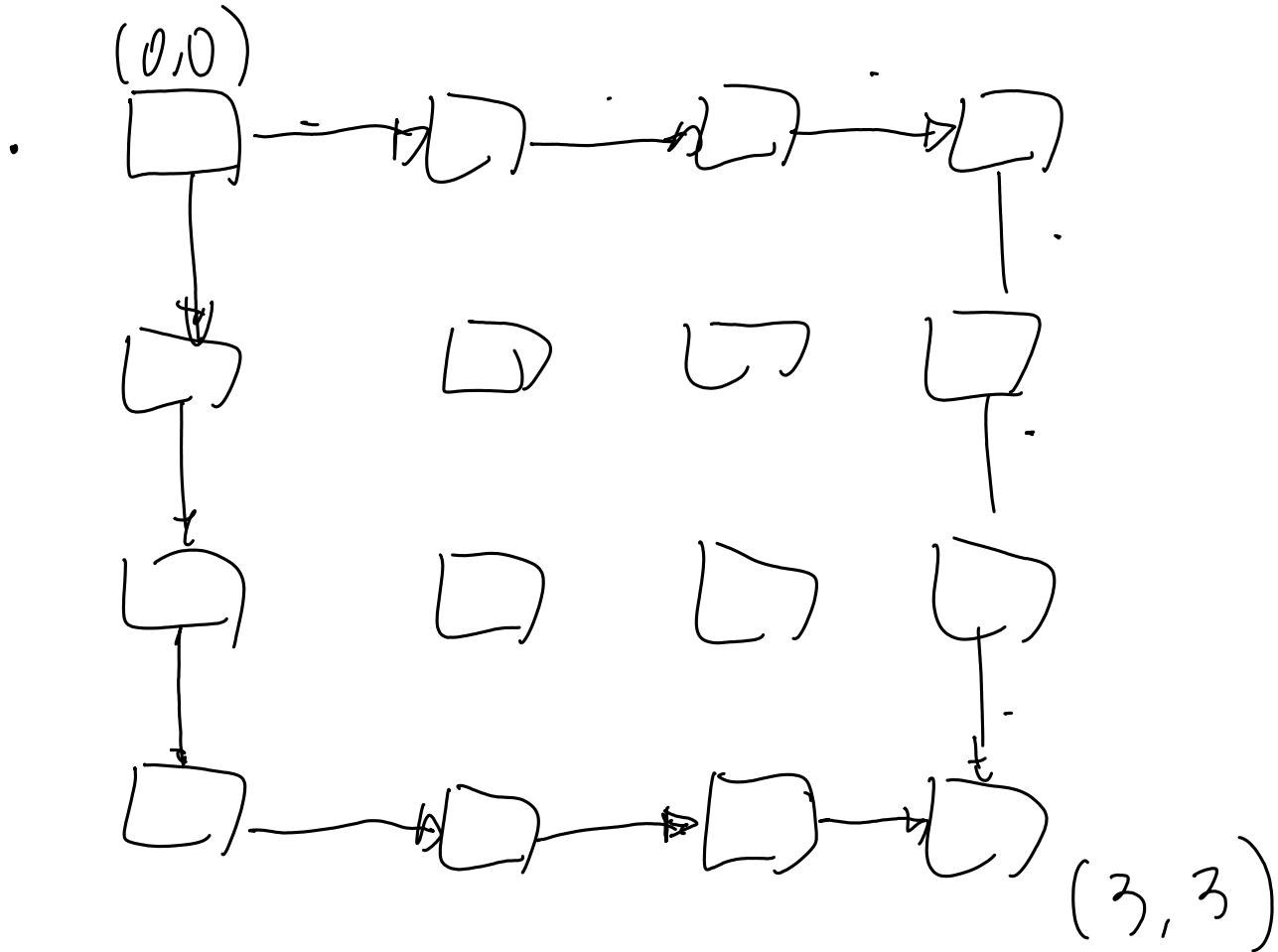
$$\approx \frac{1}{4 \cdot 10^{-3}} = (4 \cdot 10^{-3})^{-1} = \frac{1}{4} \cdot 10^3$$

$$= 250$$

$$\cdot B \geq \frac{1}{2 \cdot \frac{180.4 \text{ e } 11}{3.94 \text{ e } 14} - 3 \cdot 10^{-4}}$$

$$\approx 11$$

Question 5



• Use $\nrightarrow$ (4,4) slice has no wraparound.

• Assume $T_{\text{comms, hop}} = \alpha + \frac{D}{BW}$
for one hop.

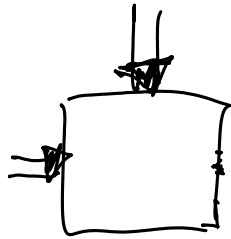
, Then, assuming pipelining:

$$T_{\text{comm}} = \# \text{Hops} \cdot \alpha + \frac{D}{BW}$$

$$\alpha = 1 \mu s$$

• $\# \text{Hops} = 6 \Rightarrow$ First byte arrives after $6 \mu s$.

• Different algos, but a lower bound on time is.



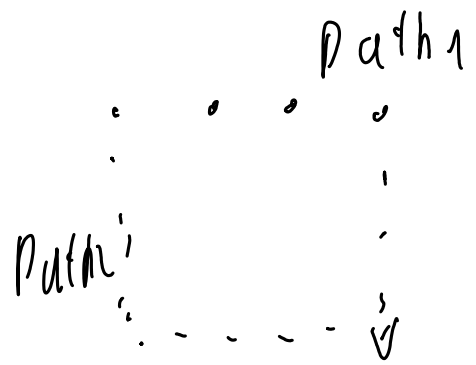
At $2 \cdot BW$

• And it needs to receive D bytes

$$\Rightarrow T \approx \frac{D}{2 \cdot BW}$$

Algo 1:

$$\frac{2 \times 8 \times 128 \times 8192}{2}$$



⇒

$$\max \left(\frac{D \text{ (Bytes)}}{2 \cdot BW_{\text{path } 1}}, \frac{D \text{ (Bytes)}}{2 \cdot BW_{\text{path } 2}} \right)$$

$$= \frac{D}{2 \cdot BW} \geq \text{Lower Bound}$$

Algo 2:

• Every thing through path 1:

$$\max \left(\frac{D}{BW_{\text{path } 1}}, 0 \right) = \frac{D}{BW_{\text{path } 1}} \geq \text{Lower Bound}$$

$$BW = 4.5 \times 10^9$$

$$\frac{D}{2} = \frac{2 \times 8 \times 128 \times 8192}{2}$$

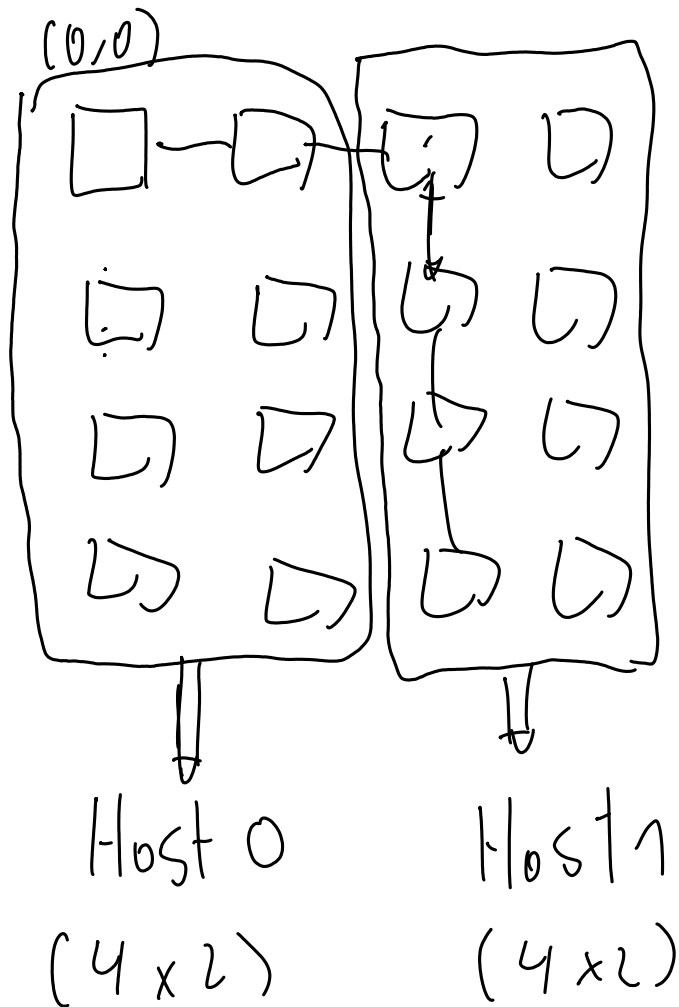
$$= 2^3 \cdot 2^7 \cdot 2^{13} = 2^{23}$$

$$\Rightarrow \frac{2^{23}}{4.5 \times 10^{10}} \approx 0.00018$$

$$= 0.18 \cdot 10^{-3} \text{ (s)}$$

$$T_{\text{ugrms}} = 6 \text{ ns} + 0.18 \text{ ms.}$$

• Question 6:



• No wraparound links

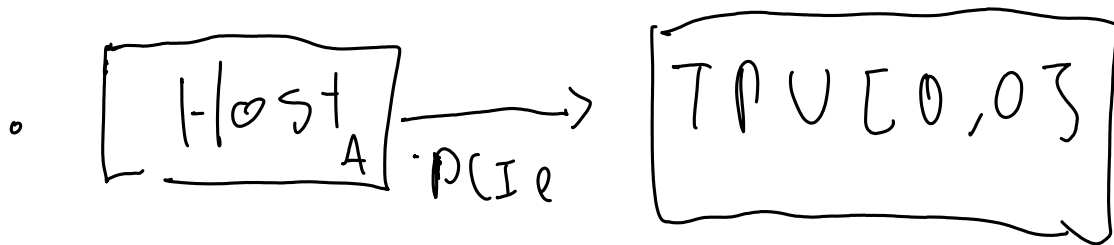
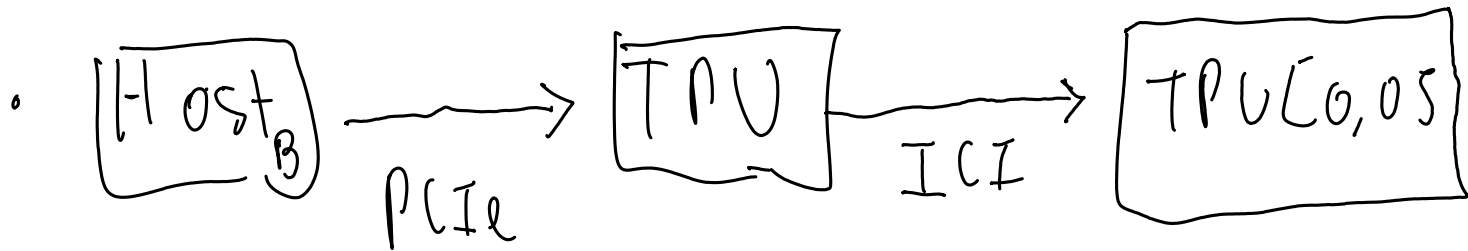
• $\text{int8 } A [128 \cdot 1024, 128 \cdot 1024]$

• Assume each host DRAM has, an even part of the array A:

$$\Rightarrow \frac{128^2 \cdot 1024^2}{2} \approx 8.6 \text{ GB}$$

per host, so it fits in v8e HBM
which is 16 GB

Path:



- Assuming pipelining: $T_{\text{comms}}(A) :=$

$$= \frac{8.6 \text{ GB}}{BW_{\text{PCIe}}} = \frac{8.6 \cdot 10^9}{1.6 \cdot 10^{10}} \approx 5.3 \cdot 10^{-1} \text{ s} = 0.53 \text{ s}$$

$$\text{bf } 16 [8, 128 * 1024]$$

$$\cdot 8 \times 128 \times 1024 \times 2 \approx 260$$

so lets assume its in HBM already.

Assuming intended operation is:

$$[128 \times 1024, 128 \times 1024] @ [128 \times 1024, 8]$$

$$\Rightarrow T = \max(T_{\text{comms}}, T_{\text{math}})$$

$$= \max\left(0.53(s), \frac{(128 \cdot 1024)^2 \cdot 8 \cdot 2}{1.47e14}\right)$$

$$\approx \max(0.53(s), 0.001s)$$

$$= 0.53s$$

